

Muhammad Ahmad

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EDUCATION

University of British Columbia

Bachelor of Applied Science in Electrical Engineering

Vancouver, BC

Expected Graduation: 2029

- Cumulative GPA: 3.85
- Relevant Coursework:** Circuit Analysis, Linear Systems, Data Structures & Algorithms, Computing Systems, Integral Calculus, Multivariable Calculus, Differential Equations, Systems & Signals, Electromagnetism

EXPERIENCE

Embedded Software Test Engineer

Stryker

May 2026 – Present

Burnaby, BC

- Engineered modules for a **C#** and **C++** Hardware-in-the-Loop (HIL) automation framework to functionally test medical light sources, increasing test coverage by 10% and reducing manual execution time by 50%
- Contributed to a Continuous Integration (CI) testing pipeline utilizing **Python**, **Bash**, and **Power Automate** to automatically flash remote builds, execute nightly integration tests, and upload validation reports
- Executed hardware electrical verification testing using **oscilloscopes**, **function generators**, and **multimeters**, analyzing signal integrity to ensure compliance with medical device reliability standards

Applied AI Team Lead

UBC Agrobot

September 2025 – Present

Vancouver, BC

- Directed a **10-member** cross-functional engineering team in designing and deploying end-to-end **Machine Learning (ML)** perception pipelines for a fully autonomous agricultural rover
- Led the development and training of **Computer Vision (CV)** architectures, including **PyTorch U-Net** segmentation models, to perform real-time **object detection**, **image classification**, and **semantic segmentation** for crop analysis

Research Assistant

Yankai Group

September 2025 – April 2026

Vancouver, BC

- Conducted experimental research on **Parameter-Efficient Fine-Tuning (PEFT)** methods (LoRA, PiSSA) on **Vision Transformer (ViT)** models, quantifying catastrophic forgetting across 5 distinct classification tasks
- Engineered a reproducible benchmarking pipeline using **Python**, **Bash**, and **Hugging Face** to systematically evaluate model retention performance and catastrophic forgetting across datasets
- Analyzed experimental outputs to identify statistical anomalies, synthesizing findings into actionable insights to guide graduate researchers in follow-up **fine-tuning** iterations

PROJECTS

Magnetic Field-Controlled RC Car | C, PIC32, STM32, EFM8, Circuit Design, I2C

- Architected a multi-microcontroller vehicle system in **C** using a **PIC32** and **EFM8** for core logic, integrating an **STM32** to process real-time joystick inputs for manual override, **cruise control**, and autonomous operating states
- Designed and built custom analog circuitry from scratch, featuring discrete **H-bridges** for motor actuation, audio feedback, and **inductor sensor arrays** to dynamically detect charge lines for autonomous magnetic-field path tracking
- Interfaced a **VL53L0C time-of-flight** sensor via **I2C** to establish a real-time **collision detection** system, automatically overriding motor drive signals to safely halt the vehicle during autonomous operations

Multi-Protocol Digital Logic Analyzer | SystemVerilog, Altera MAX 10 FPGA, Questa, SPI, I2C, UART

- Architected a digital logic analyzer on an **Altera MAX 10 (DE10-Lite)** FPGA using **SystemVerilog** to capture, decode, and analyze high-speed serial data
- Developed and integrated configurable hardware modules, including baud rate generators and frame decoders, to support multiple communication protocols including **SPI**, **I2C**, and **UART**
- Designed a robust memory architecture utilizing **FIFO buffers** to reliably capture, store, and manage decoded protocol frames across asynchronous clock domains for subsequent readout
- Rigorously verified and debugged complex logic designs by developing comprehensive testbenches in the **Questa simulation environment** prior to hardware deployment

Reflow Oven Controller | CV-8052, Assembly, Python, SSR, K-Type Thermocouple

- Engineered a closed-loop thermal control system using a **CV-8052** microcontroller, **K-type thermocouple**, and solid-state relay to execute multi-stage thermal profiles with a maximum error of 2.998°C
- Programmed a 6-stage finite state machine and a 1 ms interrupt-driven scheduler in **Assembly** for non-blocking PWM generation, digital sensor filtering, and real-time safety aborts
- Built a **Python** web dashboard to parse serial telemetry, visualize live reflow curves, and automatically generate and email PDF reports upon cycle completion

TECHNICAL SKILLS

Engineering & Hardware: PCB Design, Circuit Analysis, Signal Integrity, Microcontrollers (STM32, ESP32, PIC32, CV-8052), FPGA, Hardware-in-the-Loop (HIL)

Software & Firmware: C, C++, Python, C#, Assembly, SystemVerilog, Bash, Firmware Development, PID Control, Machine Learning (PyTorch, TensorFlow)

Tools & Protocols: Oscilloscopes, Logic Analyzers, KiCad, MATLAB, I2C, SPI, UART, Git, CI/CD, Questa, Power Automate